

Evaluating Haptic and Visual Cues in Maze Navigation Tasks

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Introduction and Background

Engineers are often assigned to complete tasks (calibration, operating systems, etc.) with a certain precision and speed. Usually, these tasks accompany real-time feedback in guidance of their action to optimize performance. This feedback often comes in different modalities, which in turn affects people’s responses. This phenomenon is related to Cognitive Load Theory, stating that the action of processing different feedback types requires different mental resources [1].

In this project, a maze-solving task is used to simulate the real-time scenario of engineers thinking and conducting their work at the same time. Particularly, this project investigates the effect of visual and haptic feedback, in the forms of flashing light and tactile vibration respectively, on the speed and accuracy of the maze-solving task.

The hypothesis is as follows:

1. Haptic feedback, in the form of a vibration, will allow participants to both:
 - (a) Complete the maze in the least amount of time, thereby achieving the greatest speed, and
 - (b) Complete the maze with the least amount of time in contact with the wall, thereby achieving the highest accuracy.
2. Additionally, there will be a negative correlation between speed and accuracy, meaning that as the speed of the participant to complete the maze increases, the accuracy decreases.

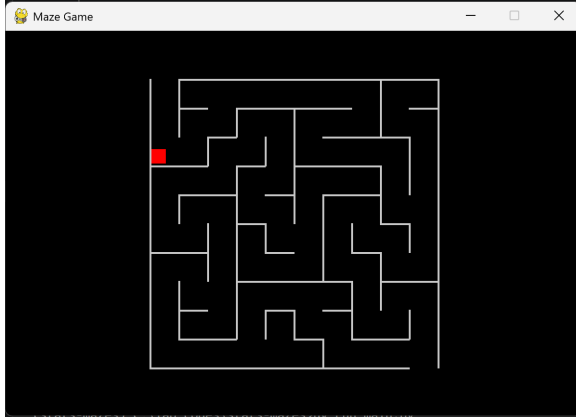
Methods

To design a maze-solving task, a 2D maze game was developed using the `pygame` library in `Python` [5]. The same maze was used in every trial to prevent differences in difficulty among participants. This maze is maneuvered by a white square cursor controlled by arrow keys on a laptop. It moves at a constant speed when the keys are held.

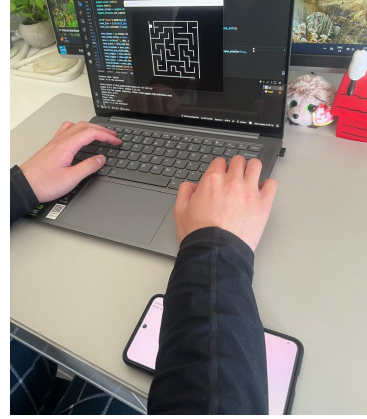
To implement the independent variable of visual feedback, the `Python` code changes the cursor color to bright red whenever the participant collides with the wall, and reverts back to light-gray otherwise. To implement the independent variable of haptic feedback, the `socket` library was used to send signals over Wi-Fi to an external device, which would then vibrate when the cursor is in contact with the wall, and stop vibrating otherwise.

When the Python code runs, 2 timers are in action:

1. A speed timer that keeps track of the total time for the participant to finish the maze. It starts when the cursor's edge enters the maze and ends when the cursor's edge exits. The time recorded is the dependent variable: Finish time, t_{speed} .
2. An accuracy timer that only runs when the cursor is in contact with the maze's wall, and pauses otherwise. This timer ends when the speed timer ends. The collision time recorded is the dependent variable: Collision time, t_{accuracy} .



(a) Visual feedback condition.



(b) Haptic feedback condition.

Figure 1: Maze interface under each experimental condition.

The two dependent variables are chosen as they are relatively simple to implement in `Python` for tracking purposes. In addition, measuring the times ensure the dependent variables are continuous.

Furthermore, two covariates were measured: hours of games played in the past 7 days, G ; and self-reported sleep quality from 1 to 10, Q . First, "gamers" are usually more familiar with maneuvering digital obstacles using keyboard controls [2]. Second, each participant would have varying degrees of focus during the test depending on their recent workload and sleep quality [3]. Tired participants may have unusually high time data, which can create outliers.

Participants

In this experiment, the population studied is Engineering Science students. This was chosen due to their similar ages, experiences and computer competency. In particular, for each of the 2 feedback types, a separate sample with size of $n = 24$ students was randomly drawn from the EngSci common room. This amounted to 48 total participants. These students were randomly selected, which prevented convenience sampling.

All participants were given the same setting and identical scripted instructions before the start of each trial. They did not know the maze structure nor the feedback type. Furthermore, each participant was given the same laptop as well as vibration and visual intensity. Only the feedback method was changed between trials, to ensure consistency.

Biases

Inevitably, bias still exists. Even though Engineering Science students were chosen for their similar technical backgrounds, one possible sampling bias would be the uneven spread of gaming experience present among them. This would create polarizing data on both ends, which this experiment does not account for.

In addition, the data transmission speed depends on the network strength, which was not always stable. This primarily affects the Haptic Feedback, since the vibration signal was sent to the phone via Wi-Fi.

Furthermore, the team observed that participants had different maze solving strategies. Some participants chose to solve the maze as they moved the cursor, while other participants chose to analyze the maze fully before moving. This created additional differences for the maze finish time.

Procedures

1. Set up the laptop on a table. Place the vibration device under the participant's arm as in Figure 1b, regardless of the actual feedback type. Start the code and enter participant name and feedback type in the main menu. Ensure the participant does not see the feedback type or maze structure beforehand.
2. A pre-screening survey was administered to evaluate each participant's prior gaming experience and current fatigue levels. Participants rated their sleep quality on a 10-point scale, with higher scores indicating a greater quality of sleep; they also report their total gaming hours in the past 7 days.
3. Inform the participants the controls and instruct them to reach the end as quick as possible while trying to avoid touching the walls as much as possible.
4. Once the participants are ready, tell them to click the START button. The finish timer starts immediately, the finish timer ends automatically when the cursor reaches the exit.

5. Wait for the participants to complete their trials and ensure that they are not disturbed throughout; otherwise, discard the results. To ensure consistency, that participant can no longer be part of any trials.

Analysis

All analysis was done using the R [4] code in Appendix A. First, Shapiro-Wilk test for normality was conducted for each dependent variable for each feedback type. Here, the Null Hypothesis is $H_0 : X \sim N(\mu, \sigma^2)$, and the Alternative Hypothesis is $H_1 : X \not\sim N(\mu, \sigma^2)$.

It was determined that the data collected was not normally distributed. In addition, the right-skewness and non-symmetry of the data also made parametric tests not applicable. The results section gives statistical evidence. Therefore, neither the f-test nor the t-test could be applied to the collected data to determine mean differences and variance ratios. The Pearson Correlation test also could not be applied to find the correlation between speed and accuracy.

Consequently, to test the difference between the means, nonparametric one-tailed Mann-Whitney U tests were conducted [6]. Two separate tests, comparing the data between Visual and Haptic feedback, were conducted for each of t_{speed} and t_{accuracy} . The Null Hypothesis for both is $H_0 : \mu_{t,\text{visual}} = \mu_{t,\text{haptic}}$. To test if Haptic feedback produces lesser finish times and lesser collision times in accordance with Hypothesis 1, the Alternative Hypothesis is $H_1 : \mu_{t,\text{haptic}} < \mu_{t,\text{visual}}$.

To test the correlation between speed and accuracy, nonparametric one-tailed Spearman's Rank Correlation tests were conducted [7]. Two separate tests were conducted, comparing the t_{speed} and t_{accuracy} within each of the Visual and Haptic feedback groups. The Null Hypothesis is $H_0 : \rho_s = 0$. To test if Speed and Accuracy are negatively correlated in accordance with Hypothesis 2, the Alternative Hypothesis is $H_1 : \rho_s < 0$. ρ_s is Spearman's Rank Correlation coefficient, with $\rho_s = 0$ denoting no correlation; $\rho_s > 0$ indicates a positive correlation; $\rho_s < 0$ indicates a negative correlation.

In addition, to determine the effects of gaming experience, G , and sleep quality, Q , on both speed and accuracy. Further one-tailed Spearman's Rank Correlation tests were conducted. Four separate tests were conducted, comparing Q, G with $t_{\text{accuracy}}, t_{\text{speed}}$ for both feedback groups. The Null Hypothesis is $H_0 : \rho_s = 0$. The Alternative Hypothesis is $H_1 : \rho_s < 0$, to investigate the effect of better sleep and more gaming experience on greater speed and better accuracy.

Results

Mean and Variance

Primary Variables

Feedback Type	Mean		Variance	
	t_{speed}	t_{accuracy}	t_{speed}	t_{accuracy}
Haptic	23.45	4.00	33.07	5.38
Visual	26.89	2.61	61.70	4.81

Table 1: Mean and Variance of Task Performance by Feedback Type for Primary Variables

Covariates

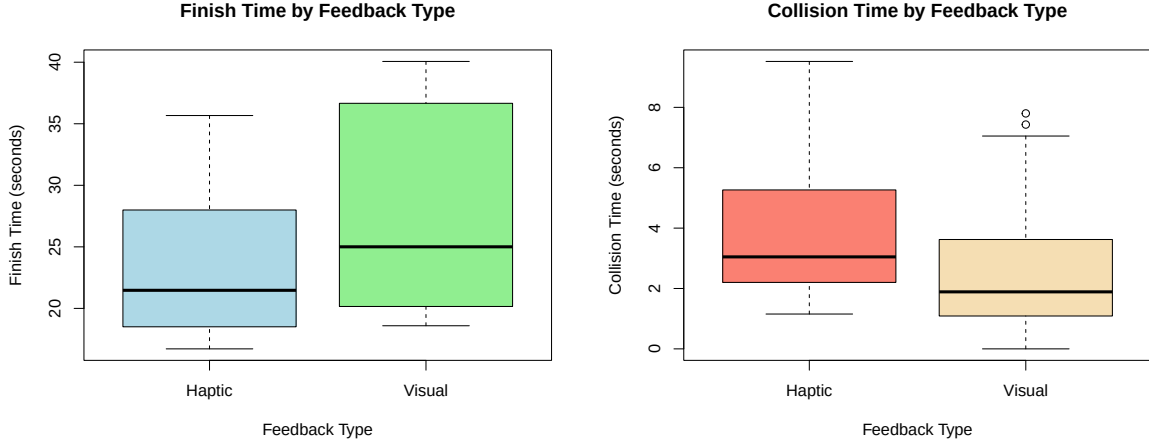
Feedback Type	Mean		Variance	
	Gaming, G	Sleep, Q	Gaming, G	Sleep, Q
Haptic	1.50	6.63	1.13	3.03
Visual	1.11	6.42	0.55	2.43

Table 2: Mean and Variance of Task Performance by Feedback Type for Covariates

Descriptive statistics were calculated to summarize task performance (Table 1). Through preliminary inspection, the Haptic group on average had a faster finish time (greater speed), while the Visual group had less collisions (greater accuracy). Statistical significance will be tested later.

Descriptive statistics were also calculated for the two measured covariates (Table 2): weekly gaming hours and sleep quality. On average, the groups demonstrated relatively balanced gaming hours and sleep quality. By tracking these metrics, any variations in task performance can be contextualized against participant fatigue and prior keyboard experience.

Data Distribution



(a) Box plot of finish time for both feedback. (b) Box plot of collision time for both feedback.

Figure 2: Box plots for each dependent variable.

The four categories of data were visualized according to the box plots in Figure 2. Through preliminary inspection of the medians and whiskers, all four groups are right-skewed, since the top whisker is longer and the medians are closer to the bottom whiskers. This means that the data is concentrated toward lower values of t_{speed} and t_{accuracy} . This gives evidence that the data does not follow a normal distribution and is asymmetric.

Furthermore, there are no significant outliers in the data, other than two points in the visual collision data 2b.

Normality Tests

Feedback	Variable	Statistic (W)	p -value	Conclusion
Visual	t_{speed}	0.83	0.0011	Non-Normal
Visual	t_{accuracy}	0.83	0.00087	Non-Normal
Haptic	t_{speed}	0.89	0.011	Non-Normal
Haptic	t_{accuracy}	0.86	0.0037	Non-Normal

Table 3: Shapiro-Wilk Normality Test Results

The results of the Shapiro-Wilk test for each group were shown in Table 3. All four groups had $p \ll 0.05$, showing that their distributions are significantly different than a normal distribution. This confirmed the Alternative Hypothesis, $H_1 : X \not\sim N(\mu, \sigma^2)$.

Mann-Whitney U Tests

Variable	Statistic (W)	p -value	Outcome
t_{speed}	372	0.0426	Reject H_0
t_{accuracy}	160	0.996	Fail to Reject H_0

Table 4: Mann-Whitney U Test Results (Alternative: Haptic < Visual)

Because the data violated the assumption of normality, the non-parametric Mann-Whitney U test was selected instead of an independent two-sample t-test.

The results of the two Mann-Whitney U Tests were given in Table 4. Values of $p < 0.05$ signified a rejection of the Null Hypothesis, meaning that a certain time for the Haptic group is less than that for the Haptic group.

For finish time (t_{speed}), the Null Hypothesis was rejected. This significantly confirmed the Alternative Hypothesis, that Haptic feedback resulted in a lesser finish time (equivalent to greater speed). This confirmed Hypothesis 1a.

For collision time (t_{accuracy}), the Null Hypothesis failed to be rejected. There was absolutely no statistical evidence that Haptic feedback had lesser collision times than Visual feedback. This strongly refutes Hypothesis 1b.

Trade-off Analysis

Primary Variables

To formally evaluate the speed-accuracy trade-off without relying on normal distribution assumptions, a Spearman’s Rank-Order Correlation was performed instead of a Pearson correlation.

Feedback	Statistic (S)	p -value	ρ_s	Outcome
Visual	2870	0.122	-0.248	Fail to Reject H_0
Haptic	2270	0.524	0.0130	Fail to Reject H_0

Table 5: Spearman’s Correlation results for primary variables (Alternative: $\rho_s < 0$)

The results of the two Spearman’s Rank-Order Correlation tests were given in Table 5. Values of ρ_s represented the strength of the correlation, while values of $p < 0.05$ signified a rejection of the Null Hypothesis. For the speed and accuracy to be negatively correlated (i.e., as speed increases, accuracy decreases), both $p < 0.05$ and $\rho_s < 0$ must be true.

For the Visual feedback group, a weak negative correlation was observed ($\rho_s = -0.248$), but this relationship was not statistically significant ($p = 0.122$). Similarly, for the Haptic feedback group, no meaningful correlation was observed ($\rho_s = 0.013$), and the result was also not statistically significant ($p = 0.524$).

Because both p-values exceed the 0.05 threshold, the Null Hypothesis failed to be rejected for both conditions. The statistical evidence indicates there was no significant speed-accuracy trade-off in this experiment, directly **refuting Hypothesis 2**. Participants' finish speed did not reliably predict their accuracy, regardless of the type of feedback they received.

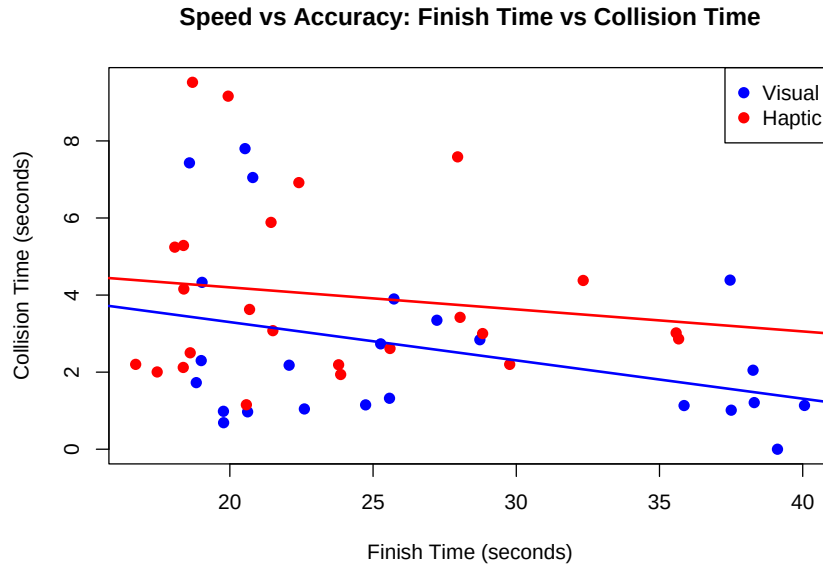


Figure 3: Scatter plot for finish time vs collision time plotted for each data point, and color-coded to differentiate Visual and Haptic groups.

A visual inspection of the scatter plot in Figure 3 supported the statistical findings that there is no correlation between finish time and collision time. The data points for both the haptic and visual groups are randomly dispersed rather than forming a downward diagonal trend (from top-left to bottom-right). This confirms the absence of a distinct speed-accuracy trade-off, showing that fast finish times was not correlated with high collision times.

Covariates

Covariate	Variable	Statistic (S)	p -value	ρ_s	Outcome
Gaming	t_{speed}	25989	0.00375	-0.411	Reject H_0
Gaming	t_{accuracy}	22550	0.126	-0.224	Fail to Reject H_0
Sleep	t_{speed}	22887	0.0971	-0.242	Fail to Reject H_0
Sleep	t_{accuracy}	22077	0.177	-0.198	Fail to Reject H_0

Table 6: Spearman’s Rank-Order Correlation Results for covariates (Alternative: $\rho_s < 0$)

The results of the four Spearman’s Rank-Order Correlation tests for the two covariates are given in Table 6. For hours spent gaming and the maze finish time, $p \ll 0.05$ and $\rho_s < 0$, rejecting the Null Hypothesis and showing a significant strong negative correlation. This presents strong statistical evidence that participants with greater gaming experience completed the maze significantly faster.

Conversely, the other three comparisons had $p \geq 0.05$, indicating that these baseline factors did not reliably predict participant times.

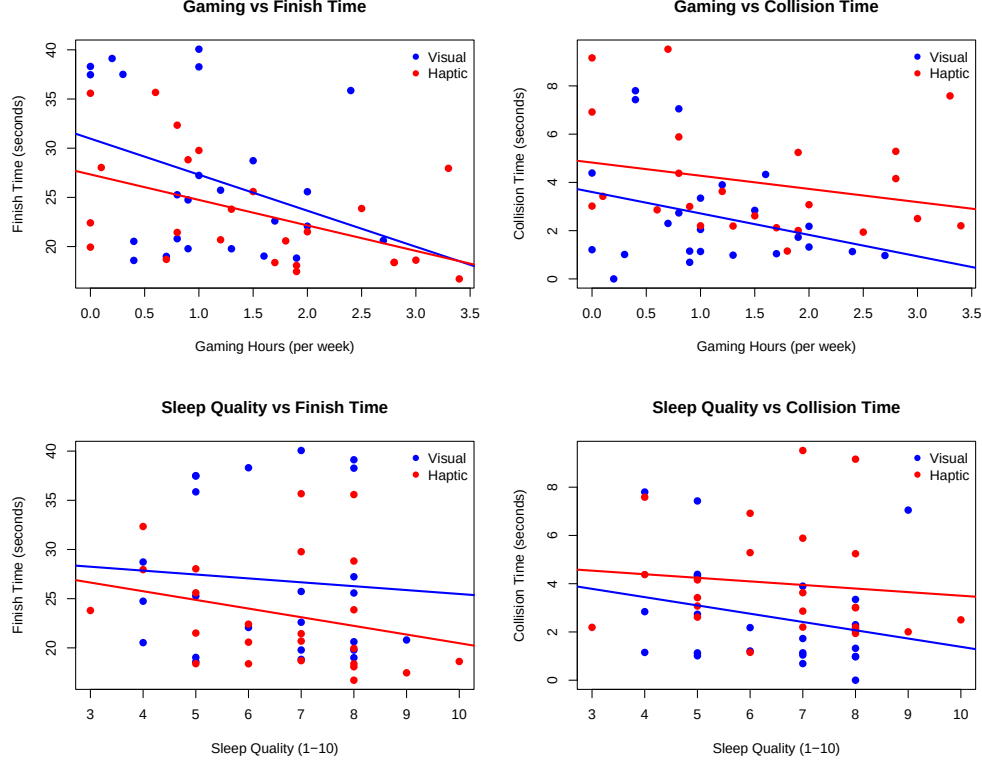


Figure 4: Scatter plots for each covariate with each time metric.

A visual inspection of the scatter plot in Figure 4 supported the statistical findings that there is a strong negative correlation between hours spent gaming and finish time, and no correlation in any other comparison.

Conclusion

Overall, the experiment explored the effects of feedback modality, both haptic and visual, on maze navigation performance, with evaluations on both finish time and collision time with maze walls. The results provided partial agreement for the proposed hypotheses and demonstrated key insights on feedback type significance in cognitively demanding tasks.

There is strong statistical evidence that haptic feedback improves task speed. The Mann-Whitney U test showed that participants using haptic feedback completed the maze faster than those using visual feedback, confirming Hypothesis 1a. However, no significant improvement in accuracy was observed, as collision time results failed to reject the null hypothesis. Thus, Hypothesis 1b is not supported. Contrary to expectations, no statistically significant speed-accuracy trade-off was found under either condition. Therefore, Hypothesis 2 is not supported, and the two metrics appear to be largely independent.

Several limitations should be acknowledged. The relatively homogeneous sample of Engineering Science students limits generalizability. Additionally, prior gaming experience may have introduced variation in keyboard proficiency, influencing finish time independently of feedback type. Similarly, sleep quality likely affected cognitive load capacity, which may explain the lack of statistical significance in collision avoidance. Future work should involve a larger and more diverse sample to improve generalizability and reduce the impact of uncontrolled covariates. In terms of experimental design, implementing the haptic feedback system on a dedicated microcontroller would improve timing precision, enabling near-instantaneous feedback. Additionally, integrating computer vision tools such as OpenCV to track eye movement could provide quantitative insight into visual attention and distraction.

In conclusion, the study demonstrates that feedback modality significantly influences task speed but not necessarily accuracy and does not provide evidence for a speed-accuracy trade-off. The findings reinforce the significance of carefully selecting feedback mechanisms in engineering and interface design, particularly when optimizing for specific performance.

References

- [1] J. Sweller, “Cognitive load during problem solving: Effects on learning,” *Cognitive Science*, vol. 12, no. 2, pp. 257–285, 1988.
- [2] L. Martinez, M. Gimenes, and E. Lambert, “Video games and board games: Effects of playing practice on cognition,” *PLoS ONE*, vol. 18, no. 3, p. e0283654, 2023.
- [3] M. Engle-Friedman *et al.*, “The role of sleep deprivation and fatigue in the perception of task difficulty and use of heuristics,” *Sleep Science*, vol. 11, no. 2, pp. 74–84, 2018.
- [4] R Core Team, “R: A Language and Environment for Statistical Computing.”, 2025.
- [5] Python Core Team, “Python: A dynamic, open source programming language.”, 2026.
- [6] H. B. Mann and D. R. Whitney, “On a test of whether one of two random variables is stochastically larger than the other,” *The Annals of Mathematical Statistics*, vol. 18, no. 1, pp. 50–60, 1947.
- [7] C. Spearman, “The proof and measurement of association between two things,” *The American Journal of Psychology*, vol. 15, no. 1, pp. 72–101, 1904.

Appendix

A R Code

```
1 data <- read.csv("maze_data_final.csv", strip.white = TRUE)
2
3 # Split the dataset immediately for targeted testing
4 visual_data <- subset(data, feedback_type == "visual")
5 haptic_data <- subset(data, feedback_type == "haptic")
6
7 # =====
8 # 2. Mean and Variance
9 # =====
10
11 # Primary variables
12 cat("\n--- Means ---\n")
13 means <- aggregate(cbind(finish_time, collision_time) ~ feedback_type,
14                     data = data, FUN = mean)
15 print(means)
16
17 cat("\n--- Variances ---\n")
18 variances <- aggregate(cbind(finish_time, collision_time) ~ feedback_type,
19                         data = data, FUN = var)
20 print(variances)
21
22 # Covariates
23 cat("\n--- Covariate Means by Feedback Type ---\n")
24 covariate_means <- aggregate(cbind(hours_gaming, sleep_quality) ~ feedback_
25                               _type,
26                               data = data, FUN = mean)
27 print(covariate_means)
28
29 covariate_vars <- aggregate(cbind(hours_gaming, sleep_quality) ~ feedback_
30                               type,
31                               data = data, FUN = var)
32
33 cat("\n--- Covariate Variances by Feedback Type ---\n")
34 print(covariate_vars)
35
36 # =====
37 # 3. Assumption Checks: Normality (Shapiro-Wilk & Q-Q Plots)
38 # =====
39
40 cat("\n--- Shapiro-Wilk Normality Test: Visual Data ---\n")
```

```

38 shapiro.test(visual_data$finish_time)
39 shapiro.test(visual_data$collision_time)
40
41 cat("\n--- Shapiro-Wilk Normality Test: Haptic Data ---\n")
42 shapiro.test(haptic_data$finish_time)
43 shapiro.test(haptic_data$collision_time)
44
45 # Visualizing Normality with Q-Q Plots
46 pdf("normality_qq_plots.pdf", width = 8, height = 8)
47
48 par(mfrow = c(2, 2))
49 qqnorm(visual_data$finish_time, main = "Q-Q Plot: Visual Finish Time")
50 qqline(visual_data$finish_time, col = "red", lwd = 2)
51
52 qqnorm(haptic_data$finish_time, main = "Q-Q Plot: Haptic Finish Time")
53 qqline(haptic_data$finish_time, col = "red", lwd = 2)
54
55 qqnorm(visual_data$collision_time, main = "Q-Q Plot: Visual Collision")
56 qqline(visual_data$collision_time, col = "blue", lwd = 2)
57
58 qqnorm(haptic_data$collision_time, main = "Q-Q Plot: Haptic Collision")
59 qqline(haptic_data$collision_time, col = "blue", lwd = 2)
60
61 par(mfrow = c(1, 1))
62 dev.off()
63
64
65 # =====
66 # 4. Mann-Whitney U Tests (Hypothesis Testing: Visual vs. Haptic)
67 # =====
68
69 # -- 1. Finish Time (Speed) --
70 cat("\n--- Mann-Whitney U Test: Finish Time (Alternative: Haptic < Visual)
    ---\n")
71 wilcox.test(haptic_data$finish_time, visual_data$finish_time, alternative
    = "less", exact = FALSE)
72
73 pdf("finish_time_boxplot.pdf", width = 6, height = 5)
74 boxplot(finish_time ~ feedback_type, data = data,
75         main = "Finish Time by Feedback Type",
76         xlab = "Feedback Type",
77         ylab = "Finish Time (seconds)",
78         col = c("lightblue", "lightgreen"),
79         names = c("Haptic", "Visual"))

```

```

80 dev.off()
81
82 # -- 2. Collision Time (Accuracy) --
83 cat("\n--- Mann-Whitney U Test: Collision Time (Alternative: Haptic <
      Visual) ---\n")
84 wilcox.test(haptic_data$collision_time, visual_data$collision_time,
      alternative = "less", exact = FALSE)
85
86 pdf("collision_time_boxplot.pdf", width = 6, height = 5)
87 boxplot(collision_time ~ feedback_type, data = data,
88         main = "Collision Time by Feedback Type",
89         xlab = "Feedback Type",
90         ylab = "Collision Time (seconds)",
91         col = c("salmon", "wheat"),
92         names = c("Haptic", "Visual"))
93 dev.off()
94
95
96 # =====
97 # 5. Correlation Visualized (Scatter Plot)
98 # =====
99 # Saving a combined scatter plot for both groups
100 pdf("scatter_plot_correlation.pdf", width = 7, height = 5)
101
102 # Plotting the raw dots (Blue for Visual, Red for Haptic)
103 plot(data$finish_time, data$collision_time,
104      main = "Speed vs Accuracy: Finish Time vs Collision Time",
105      xlab = "Finish Time (seconds)",
106      ylab = "Collision Time (seconds)",
107      col = ifelse(data$feedback_type == "visual", "blue", "red"),
108      pch = 16, cex = 1.2) # pch=16 makes the dots solid circles
109
110 # Adding a legend so we know which color is which
111 legend("topright", legend = c("Visual", "Haptic"),
112      col = c("blue", "red"), pch = 16, pt.cex = 1.2)
113
114 # Adding trend lines (lines of best fit) for both groups
115 abline(lm(collision_time ~ finish_time, data = visual_data), col = "blue",
116      lwd = 2)
116 abline(lm(collision_time ~ finish_time, data = haptic_data), col = "red",
117      lwd = 2)
117
118 dev.off()
119

```

```

120
121 # =====
122 # 6. Spearman Rank Correlation (Speed vs. Accuracy Tradeoff)
123 # =====
124
125 cat("\n--- Spearman Correlation: Visual Data (Speed vs Accuracy) ---\n")
126 cor.test(visual_data$finish_time, visual_data$collision_time, method = "
    spearman", alternative = "less", exact = FALSE)
127
128 cat("\n--- Spearman Correlation: Haptic Data (Speed vs Accuracy) ---\n")
129 cor.test(haptic_data$finish_time, haptic_data$collision_time, method = "
    spearman", alternative = "less", exact = FALSE)
130
131 # =====
132 # 7. Spearman Correlation for Covariates
133 # =====
134
135 cat("\n--- Spearman Correlation: Gaming Hours vs. Finish Time ---\n")
136 # Testing if more gaming equals faster completion (negative correlation)
137 cor.test(data$hours_gaming, data$finish_time,
138         method = "spearman",
139         exact = FALSE)
140
141 cat("\n--- Spearman Correlation: Gaming Hours vs. Collision Time ---\n")
142 # Testing if more gaming equals fewer collisions (negative correlation)
143 cor.test(data$hours_gaming, data$collision_time,
144         method = "spearman",
145         exact = FALSE)
146
147 cat("\n--- Spearman Correlation: Sleep Quality vs. Finish Time ---\n")
148 # Testing if better sleep equals faster completion (negative correlation)
149 cor.test(data$sleep_quality, data$finish_time,
150         method = "spearman",
151         exact = FALSE)
152
153 cat("\n--- Spearman Correlation: Sleep Quality vs. Collision Time ---\n")
154 # Testing if better sleep equals fewer collisions (negative correlation)
155 cor.test(data$sleep_quality, data$collision_time,
156         method = "spearman",
157         exact = FALSE)

```

B Raw Data

Table 7: Experimental Results: Finish Times, Collision Times, and Covariates (Trials 1–48)

Trial	Feedback Type	Finish Time, t_{speed} (s)	Collision Time, $t_{accuracy}$ (s)	Gaming Hrs, G	Sleep Quality, Q
1	visual	18.59	7.43	0.4	5
2	visual	20.80	7.05	0.8	9
3	visual	20.53	7.80	0.4	4
4	visual	19.78	0.98	1.3	8
5	visual	35.86	1.13	2.4	5
6	visual	22.60	1.04	1.7	7
7	visual	19.00	2.30	0.7	8
8	visual	20.62	0.97	2.7	8
9	visual	22.07	2.18	2.0	6
10	visual	19.03	4.33	1.6	5
11	visual	25.57	1.32	2.0	8
12	visual	19.78	0.69	0.9	7
13	visual	37.47	4.39	0.0	5
14	visual	38.31	1.21	0.0	6
15	visual	18.83	1.73	1.9	7
16	visual	25.73	3.90	1.2	7
17	visual	37.50	1.01	0.3	5
18	visual	25.27	2.73	0.8	5
19	visual	40.06	1.14	1.0	7
20	visual	39.12	0.00	0.2	8
21	visual	38.27	2.05	1.0	8
22	visual	28.73	2.84	1.5	4
23	visual	24.74	1.15	0.9	4
24	visual	27.23	3.35	1.0	8
25	haptic	18.37	2.12	1.7	8
26	haptic	18.38	5.29	2.8	6
27	haptic	19.94	9.16	0.0	8
28	haptic	20.69	3.63	1.2	7
29	haptic	28.82	3.00	0.9	8
30	haptic	20.58	1.15	1.8	6
31	haptic	18.62	2.50	3.0	10
32	haptic	25.59	2.61	1.5	5
33	haptic	23.87	1.94	2.5	8
34	haptic	35.67	2.86	0.6	7
35	haptic	32.34	4.38	0.8	4
36	haptic	35.58	3.02	0.0	8
37	haptic	18.39	4.16	2.8	5
38	haptic	16.71	2.20	3.4	8

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Table 7 continued from previous page

Trial	Feedback Type	Finish Time, t_{speed} (s)	Collision Time, $t_{accuracy}$ (s)	Gaming Hrs, G	Sleep Quality, Q
39	haptic	22.41	6.92	0.0	6
40	haptic	29.77	2.20	1.0	7
41	haptic	28.04	3.42	0.1	5
42	haptic	21.50	3.08	2.0	5
43	haptic	23.80	2.19	1.3	3
44	haptic	18.08	5.24	1.9	8
45	haptic	17.46	2.00	1.9	9
46	haptic	27.95	7.59	3.3	4
47	haptic	18.70	9.52	0.7	7
48	haptic	21.44	5.89	0.8	7

C Q-Q Plots for Normality

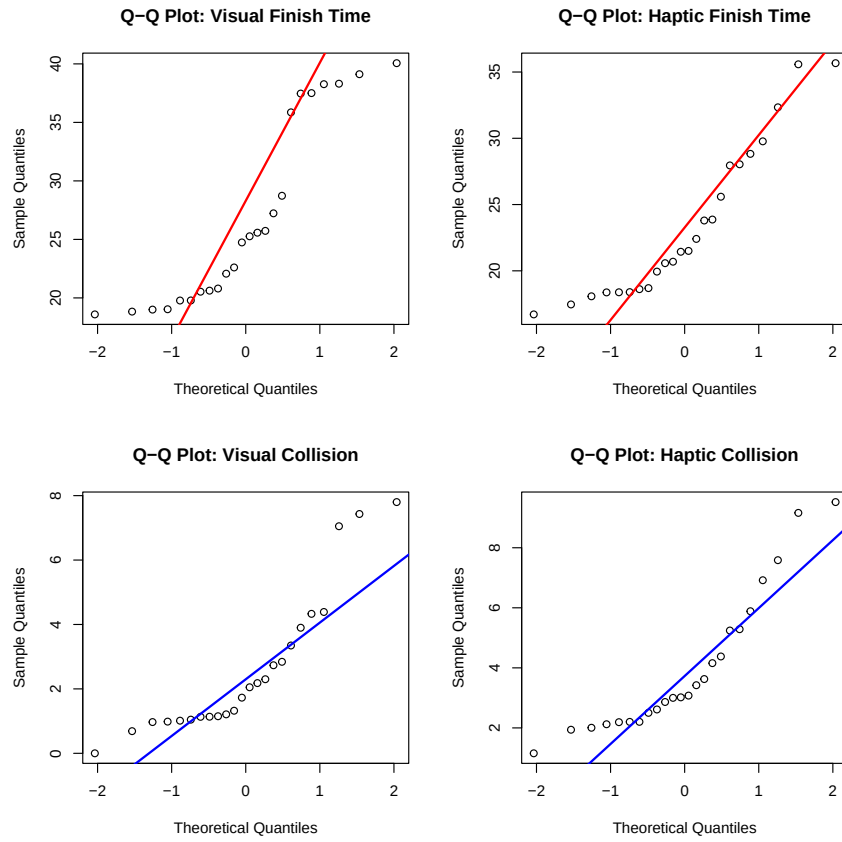


Figure 5: Normal Q-Q plots for the four groups. The points aligning with the red/blue line denote data that follows the normal distribution.

Attribution Table

This form must be filled out and signed by each team member for your project final report. The completed form must be attached to the report, though not included as part of the document (i.e., not included in the table of contents, page numbers, or word count limits). It should accurately reflect each team members' contribution to the project.

If there are irreconcilable differences that prevent all team members from signing the attribution table, then each team member must write a letter (max 800 words) explaining their position on the difference and suggest a solution. Making fraudulent claims in an attribution table displays intent to deceive and is a serious academic offence.

Contribution Matrix

Fill in abbreviations for roles for each of the required content elements. You do not have to fill in every cell.

Report Section	Rooney Cheung	Oscar Low	Oscar Liang
Introduction	WD, ET	RS, FR	RS, ET, FR
Methods - Participants	WD, CM, FR	ET, FR	ET, FR
Methods - Participants (Cont.)	WD, ET, FR	WD, ET, FR	RS, FR
Methods - Procedures	WD, EX, FR	EX, ET, FR	EX, FR
Analysis	CM, FR	WD, FR	FR
Results	DA, FR	WD, DA, FR	ET, FR
Conclusion	ET, FR	ET, FR	WD, CM, FR
Appendices	FR	ET, FR	WD, FR

Role Abbreviations

- **RS:** research
- **EX:** ran the experiment
- **DA:** performed data analysis
- **WD:** wrote first draft
- **MR:** major revision
- **ET:** edited for grammar and spelling
- **CM:** responsible for compiling elements into the complete document
- **FC:** final compilation of completed document
- **FR:** final read through for flow and consistency
- **OR:** other (OR1: _____, OR2: _____)

Declarations and Signatures

By signing below, you verify that you have:

- Read the attribution table and agree that it accurately reflects your contribution to the associated document.
- Written the sections of the document attributed to you and that they are entirely original.
- Accurately cited and referenced any ideas or expressions of ideas taken from other sources according to an accepted standard.
- Read the University of Toronto Code of Behaviour on Academic Matters and understand the definition of academic offense includes (but is not limited to) all forms of plagiarism.
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Print Name: Oscar Liang

Signature: Oscar Liang

Print Name: Oscar Low

Signature: Oscar Low

Print Name: Rooney Ming Ho Cheung

Signature: Rooney Cheung